3	**(a)**		1	**B1** **[1]**	**2.2a**	Cao Must be clear that the limit is **equal to 1**. Therefore tending to or approaching 1 or 0.999 or → 1 etc do not score
3	**(b)**		$a_n = 1 - \frac{1}{n+1}$ $n+1$ increases as n increases so $\frac{1}{n+1}$ decreases Less is being taken away from 1 each time so $a_n = 1 - \frac{1}{n+1}$ increases	**M1** **M1** **E1**	**3.1a** **2.4** **2.1**	 Convincing explanation that $\frac{1}{n+1}$ decreases Convincing completion (A.G.) **Special Case (**Max 2 for this method) Correct differentiation and states > 0 to demonstrate increasing **SC B1** Goes on to consider [positive] integers as a subset of reals to complete their argument **SC B1** dep on first **B1**.
				[3]		

			Alternative method 1 $\frac{n+1}{n+2}$ $n(n+2) < (n+1)^2$ $n(n+2) = n^2 + 2n$ and $(n+1)^2 = n^2 + 2n + 1$ Expanding brackets to give convincing completion (A.G.)	**M1** **M1** **E1**		This method is testing $\frac{n}{n+1} < \frac{n+1}{n+2}$ Formulating expression for (n+1)th or (n-1)th term For formulating inequality and cross multiplying Probably see $\frac{n+1}{n+2} > \frac{n}{n+1}$ [i.e. increasing]
			Alternative method 2 $\frac{n+1}{n+2}$ Difference is $\frac{n+1}{n+2} - \frac{n}{n+1} = \frac{(n+1)^2 - n(n+2)}{(n+1)(n+2)}$ $\frac{(n+1)^2 - n(n+2)}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)} > 0$ So sequence is increasing	**M1** **M1** **E1**		Formulating expression for (n+1)th term Finding difference and attempts to form a single fraction eg may be $a_n - a_{n+1}$ Convincing completion (A.G.)
				[3]		